

Lohith M

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with **4 years** of experience spanning the full journey from **data analytics** to **production Generative AI**, now building **agentic AI systems** at **Visa** behind Visa's **first GenAI-powered BI chatbot and enterprise developer-productivity platform**. Started in BFSI and airline analytics engineering **SQL/ETL pipelines** and **Power BI** dashboards, grew into a **Data Scientist** shipping **fraud, anomaly, and cancellation models (XGBoost, PyTorch)**, and now architect **RAG pipelines, multi-agent workflows, and LLM fine-tuning (LangChain, LangGraph, MCP)**. Skilled end-to-end across **model training, fine-tuning, REST API deployment on AWS, MLOps, and production monitoring**. Holds an **MS in Computer Science** from Auburn University at Montgomery.

TECHNICAL SKILLS

Programming & Databases: Python (Pandas, NumPy, SciPy), SQL (PostgreSQL, MySQL, Oracle), Power Query, JSON

Machine Learning: XGBoost, Scikit-learn, Optuna, Isolation Forest, Logistic Regression, Feature Engineering, Anomaly Detection, Statistical Modeling

Deep Learning & NLP: PyTorch, TensorFlow, Keras, Neural Networks (ANN, RNN), NLP, Tokenization, Embeddings

Generative & Agentic AI: RAG, Fine-Tuning (LoRA, QLoRA, PEFT), Prompt Engineering, Multi-Agent Workflows, ReAct Agents, Function Calling, Guardrails, Human-in-the-Loop, Knowledge Graphs, Vector Databases (FAISS, Pinecone, ChromaDB)

LLM & Agent Frameworks: LangChain, LangGraph, LangSmith, CrewAI, Semantic Kernel, Model Context Protocol (MCP), A2A Protocol, GPT-4, Claude, LLaMA

MLOps, Cloud & Deployment: AWS (Bedrock, SageMaker, OpenSearch, Lambda, S3, EC2), FastAPI, Flask, MLflow, Docker, Kubernetes (EKS/ECS), Terraform, CI/CD, Observability, Git

Data Engineering & BI: Apache Airflow, DBT, Amazon Redshift, ETL Pipelines, Power BI (DAX, Data Modeling), Advanced Excel, Data Governance & Quality

PROFESSIONAL WORK EXPERIENCE

AI/ML Engineer

Atlanta, GA

VISA

Aug 2024 - Present

- Developed **fraud-detection** and **anomaly-detection** models (**XGBoost, Optuna, Isolation Forest, PyTorch**) with **feature engineering** and **probability scoring** for transaction-risk and dispute propensity, improving detection precision by **12%**, and stood up **MLOps** workflows with **MLflow** tracking for reproducible training-to-deployment.
- Bridged into GenAI by **fine-tuning** foundation **LLMs (GPT-4, Claude)** for financial-domain risk and regulatory-compliance tasks using **LoRA, QLoRA**, and **PEFT**, with **prompt engineering** and **in-context learning** for reliability, and built a **model-benchmarking harness** measuring accuracy, latency, and cost to make **model selection** data-driven.
- Building on that foundation, shipped Visa's **first GenAI-powered BI chatbot** end-to-end, grounding answers on live transaction data through **RAG, vector databases**, and **function calling**, with **multi-provider** inference across **GPT, Claude**, and **LLaMA** routed for accuracy and cost.
- Extended the chatbot into daily reporting with a **Text2SQL agent** that converts natural language to **SQL**, cutting reliance on data teams for basic reporting by **40%**, and integrated **Power BI** so users saw conversational answers alongside visual transaction trends.
- Moved from single-purpose to autonomous agents: built an **Operational Assistant** with **LangGraph** and **AWS Bedrock** that auto-triages and routes tickets, and **ReAct agents** that pull real-time logs to run **root-cause analysis** via **LLMs**, cutting **MTTR**, both governed by **human-in-the-loop** gates and **guardrails**.
- Scaled from individual agents to **multi-agent workflows** using **LangGraph, CrewAI**, and **Semantic Kernel**, connecting them to enterprise systems (**Jira, Confluence, GitHub, Checkmarx, Nexus IQ, CI/CD**) through a custom **MCP**

integration layer with **PR-aware security remediation**.

- To keep the growing agent fleet discoverable, built an **MCP Registry** and **Agent-2-Agent (A2A) Registry** as a single source of truth for deployed **MCP servers** and agents, with **A2A protocol** support for cross-agent **orchestration**.
- Optimized platform economics by **routing tasks across models by complexity** (open-source, **Claude Code**, and **Codex**), trimming **GenAI operating cost by 20%** while holding quality steady, verified via the benchmarking **harness**.
- Hardened everything for production with **observability** and **AI governance: deterministic** and **LLM-as-judge graders**, **LangSmith** tracing, **audit logging**, and **circuit-breaker** guardrails, which reduced developer **code-review effort by 30%**.
- Delivered **CodeGenie**, a **RAG-based** enterprise code-and-architecture search at **95%+ accuracy**, using **embeddings retrieval** and **knowledge-graph** injection, served via **FastAPI** on **AWS (Bedrock, SageMaker, OpenSearch)**.
- Deployed the agentic platform on **Kubernetes (Amazon EKS/ECS)** with **Docker** and **Terraform**, moving services off **VMs**, and helped fellow engineers adopt it through docs and walkthroughs on **MCP** and **A2A integration**.

Data Scientist

American Airlines

Hyderabad, India

Dec 2022 - Jul 2023

- Built flight-cancellation classification models (**XGBoost, Logistic Regression**) on weather and crew data, achieving 87% accuracy and enabling proactive operational planning.
- Engineered features and preprocessing (**Pandas, NumPy, SciPy, Scikit-learn**) with **encoding, scaling, and outlier handling**, improving dataset quality and reducing training variance by **18%**.
- Validated model performance using **cross-validation** and **ROC-AUC**, tuning **classification thresholds** to balance false cancellations against missed disruptions for reliable operational decisions.
- Automated **ETL data pipelines (Pandas, Apache Airflow, AWS S3)** with scheduled **orchestration** and **data-validation checks**, consolidating **5+ operational systems** into clean, model-ready datasets.
- Analyzed boarding and delay data across 3+ hub airports (Python, SQL) to support on-time performance reporting.
- Deployed models behind a **Flask REST API** with scheduled batch scoring, and monitored **prediction drift** to keep accuracy stable as seasonal flight patterns shifted.
- Developed **Power BI** dashboards (**DAX, Amazon Redshift, DBT**) tracking load-factor and fuel-usage KPIs across 3+ hubs for real-time operational visibility.

Senior Representative, Data Analyst (BFSI)

Concentrix

Hyderabad, India

Sep 2021 - Dec 2022

- Delivered quantitative analysis and reporting for **5+ enterprise BFSI** clients across risk management, compliance monitoring, and operational decision-making.
- Migrated legacy queries to **Oracle SQL** and built Power Query **ETL** workflows, improving query performance by **25%** and data availability by **35%**.
- Designed **10+ interactive Power BI** dashboards (**DAX, data modeling**), accelerating finance-stakeholder decisions.
- Performed exploratory data analysis and feature engineering (**Python, Pandas, NumPy**) to detect anomalies across **1M+** financial records, building domain foundations for fraud modeling.
- Automated reporting (**Excel macros, Power BI Service**), reducing manual effort by **40%** and establishing data-quality and governance frameworks.

Process Associate, Real Estate Analytics

RealPage

Hyderabad, India

Jul 2021 - Sep 2021

- Cleaned and validated real-estate transaction datasets (**Excel, SQL, data-quality checks**), improving data consistency.
- Built **15+ recurring Excel and Google Sheets** reports covering leasing and operational metrics, cutting manual report-preparation time by 30% for stakeholders.

PROJECTS

DeshMate – Cultural Roommate-Matching Platform - <https://www.deshmate.com>

- Built and launched a production roommate-finder marketplace for immigrant students and professionals across 36+ communities, with Google OAuth and verified-email auth, photo listings, map-based search, saved rooms, reviews, and an admin moderation dashboard with automated removal emails.
- Engineered a zero-framework stack for speed and cost (sub-second loads, near-zero infra spends) PostgreSQL with row level security, serverless email APIs, CDN edge deploys with CI smoke tests and uptime alerting, shipping daily production releases to real users.

Tech Stack:

JavaScript (ES6), HTML/CSS, Supabase (PostgreSQL, RLS, Auth, Storage), Vercel Serverless Functions, Google OAuth 2.0, Resend API, Leaflet.js, GitHub Actions, Cloudflare DNS

Porzolio - Portfolio Builder Web App - <https://porzolio.vercel.app/>

- Built and deployed a no-code portfolio builder where users drag, resize, and style blocks (text, links, images, video, audio, PDFs, resumes) on a free-form canvas and publish them instantly as a shareable public profile, with multi-page workspaces, starter templates, and an opt-in Explore directory.
- Engineered a custom vanilla-JavaScript canvas engine (selection, drag/resize, z-ordering, undo, rich-text toolbar) bridged into React, persisted each site as a single JSON block document in Supabase Postgres with autosave, secured multi-tenant data using Row Level Security and Supabase Auth (Google OAuth, magic link), sanitized user HTML to prevent stored XSS, and enriched links via Microlink, YouTube Embed, and GitHub REST APIs.

Tech Stack: Next.js 16, React 19, Supabase (Postgres, Auth, Storage), Vanilla JS, PostgreSQL RLS, Vercel

Customer Churn Prediction - ANN Binary Classification ([GitHub](#))

- Built a TensorFlow/Keras neural network for churn prediction with feature engineering and hyperparameter tuning. Stack: Python, TensorFlow, Keras, Scikit-learn.

IMDB Sentiment Analysis - RNN Text Classification ([GitHub](#))

- Developed a SimpleRNN (Keras) for binary sentiment classification with tokenization and embedding layers. Stack: Python, TensorFlow, Keras, NLP.

EDUCATION

Master of Science, Computer Science - Auburn University at Montgomery, Montgomery, AL | 2023 - 2025

Bachelor of Technology, Electronics & Communication Engineering - Anurag University, India | 2016 - 2020

COMMUNITY & THOUGHT LEADERSHIP

- Author AI/ML technical articles on [Medium](#) and share applied machine-learning insights with a growing professional audience on LinkedIn.